

JavaScript Array Method - shift()

The `shift()` method is one of the most commonly used array methods in JavaScript. It is used to **remove the first element** from an array.

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What is shift()?

Definition

The `shift()` method removes the **first element** from an array.

It shifts all remaining elements one position to the left.

It modifies the **original array** and returns the **removed element**.

Why Use shift()?

We use `shift()` when we need to:

- Remove the first item
 - Process a queue
 - Remove the oldest notification
 - Remove the first task
 - Process API data one by one
-

Syntax

```
array.shift();
```

Parameters

`shift()` does **not** accept any parameters.

```
array.shift();
```

Return Value

`shift()` returns the **removed first element**.

Example

```
const fruits = ["Apple", "Mango", "Orange"];

let removed = fruits.shift();

console.log(removed);
```

OUTPUT

Apple

How shift() Works

Before

```
["Apple", "Mango", "Orange"]
```

↓

```
shift()
```

↓

Removed

```
Apple
```

↓

After

```
["Mango", "Orange"]
```

Notice that every remaining element moves one position to the left.

Example 1 (Basic)

```
const fruits = [  
  "Apple",  
  "Mango",  
  "Orange"  
];  
  
fruits.shift();  
  
console.log(fruits);
```

OUTPUT

```
["Mango", "Orange"]
```

EXPLANATION

- "Apple" is removed.
 - "Mango" becomes the first element.
 - "Orange" becomes the second element.
 - The original array is modified.
-

Example 2 (Store Removed Element)

```
const fruits = [  
  "Apple",  
  "Mango",  
  "Orange"  
];  
  
const removedFruit = fruits.shift();  
  
console.log(removedFruit);  
  
console.log(fruits);
```

OUTPUT

```
Apple  
  
["Mango", "Orange"]
```

Example 3 (Numbers)

```
const numbers = [  
  
  10,  
  
  20,  
  
  30,  
  
  40  
  
];  
  
numbers.shift();  
  
console.log(numbers);
```

OUTPUT

```
[20,30,40]
```

Example 4 (Objects)

```
const users = [  
  
  {  
  
    name: "John"  
  
  },  
  
  {  
  
    name: "Sara"  
  
  }  
  
];  
  
const removedUser = users.shift();  
  
console.log(removedUser);  
  
console.log(users);
```

OUTPUT

```
{ name: "John" }  
  
[  
  { name: "Sara" }  
]
```

Example 5 (Empty Array)

```
const numbers = [];  
  
const removed = numbers.shift();  
  
console.log(removed);  
  
console.log(numbers);
```

OUTPUT

```
undefined  
  
[]
```

Example 6 (Multiple shift())

```
const colors = [  
  "Red",  
  "Blue",  
  "Green"  
];  
  
colors.shift();  
  
colors.shift();  
  
console.log(colors);
```

OUTPUT

```
[ "Green" ]
```

Example 7 (Using Variables)

```
const tasks = [  
  "Study",  
  "Exercise",  
  "Sleep"  
];  
  
let firstTask = tasks.shift();  
  
console.log(firstTask);
```

OUTPUT

```
Study
```

Example 8 (Single Element Array)

```
const items = [  
  "Laptop"  
];  
  
items.shift();  
  
console.log(items);
```

OUTPUT

```
[]
```

Real Project Example 1

| Queue System

```
const queue = [  
  
  "Customer 1",  
  
  "Customer 2",  
  
  "Customer 3"  
  
];  
  
const servedCustomer = queue.shift();  
  
console.log(servedCustomer);  
  
console.log(queue);
```

OUTPUT

```
Customer 1  
  
["Customer 2","Customer 3"]
```

Real Project Example 2

| Notification System

```
const notifications = [  
  "Message",  
  "Payment",  
  "Order"  
];  
  
notifications.shift();  
  
console.log(notifications);
```

OUTPUT

```
["Payment", "Order"]
```

Real Project Example 3

| Todo List

```
const todos = [  
  "Wake Up",  
  "Study",  
  "Exercise"  
];  
  
const completedTask = todos.shift();  
  
console.log(completedTask);  
  
console.log(todos);
```

OUTPUT

```
Wake Up  
  
["Study", "Exercise"]
```

Real Project Example 4

| Playlist

```
const playlist = [  
  "Song A",  
  "Song B",  
  "Song C"  
];  
  
playlist.shift();  
  
console.log(playlist);
```

OUTPUT

```
["Song B", "Song C"]
```

Real Project Example 5

| Order Processing

```
const orders = [  
  "#1001",  
  "#1002",  
  "#1003"  
];  
  
const processingOrder = orders.shift();  
  
console.log(processingOrder);  
  
console.log(orders);
```

OUTPUT

```
#1001  
  
["#1002", "#1003"]
```

Best Practices

- ✓ Use `shift()` only when removing the **first element**.
-

- ✓ Store the removed element if it is needed later.

```
const firstItem = array.shift();
```

- ✓ Check if the array is empty before calling `shift()`.

```
if(array.length > 0){  
  
    array.shift();  
  
}
```

- ✓ Use meaningful variable names.

```
removedTask  
  
servedCustomer  
  
firstMessage
```

Common Mistakes

| Mistake 1

Expecting `shift()` to return the updated array.

Wrong

```
const result = numbers.shift();  
  
console.log(result);
```

Output

```
10
```

It returns the **removed element**, not the updated array.

Correct

```
numbers.shift();  
  
console.log(numbers);
```

Mistake 2

Passing a parameter.

Wrong

```
numbers.shift(10);
```

`shift()` ignores all parameters.

Correct

```
numbers.shift();
```

Mistake 3

Using `shift()` on an empty array.

```
const arr = [];  
  
console.log(arr.shift());
```

OUTPUT

```
undefined
```

| Mistake 4

Using the wrong method.

To remove the **last element**, use

```
pop()
```

instead of

```
shift()
```

| Mistake 5

Forgetting that indexes change after `shift()`.

Before

```
["Apple", "Mango", "Orange"]
```

After

```
shift()
```

Result

```
["Mango", "Orange"]
```

Now,

```
array[0]
```

is

```
"Mango"
```

not "Apple".

Interview Questions

| What does `shift()` do?

It removes the **first element** from an array.

| Does `shift()` modify the original array?

✓ Yes.

| What does `shift()` return?

The removed first element.

| Does `shift()` take any parameters?

✗ No.

| What happens if the array is empty?

It returns

undefined

| Which method removes the last element?

pop()

| Which method adds an element at the beginning?

unshift()

| What happens to the remaining elements after `shift()`?

They move one position to the left.

Summary Table

Feature	Description
Method	<code>shift()</code>
Purpose	Remove the first element
Parameters	None
Returns	Removed first element
Changes Original Array	✔ Yes
Removes From	Beginning of the array
Common Uses	Queue, Notifications, Todo List, Order Processing

Quick Comparison

Method	Action	Returns	Changes Original Array
<code>push()</code>	Add to end	New length	✔ Yes
<code>pop()</code>	Remove from end	Removed element	✔ Yes
<code>unshift()</code>	Add to beginning	New length	✔ Yes
<code>shift()</code>	Remove from beginning	Removed element	✔ Yes

Final Notes

The `shift()` method is commonly used in applications where the **oldest item needs to be processed first**. Typical use cases include:

- ✔ Queue systems (FIFO)
- ✔ Customer service queues

-  Notification management
-  Task processing
-  Order processing
-  Playlist management

Mastering `shift()` helps you understand queue-based data processing and prepares you for more advanced array operations.